- ***Chronology:*** Each block acts like a repository that stores information pertaining to a transaction, and links to the previous block in the same transaction. These connected blocks form a chronological chain providing a trail of the underlying transaction.
- ***Consensus based:*** A transaction on the blockchain is executed only if all the parties on the network unanimously approve it. Consensus-based rules can be altered to suit various circumstances.
- ***Digital signature:*** Blockchain enables exchange of transactional values using unique digital signatures that rely on public keys. Public keys are decryption code known to everyone on the network. Private keys are codes known only to the owner to create proof of ownership. This is very critical to avoid fraud in records management.
- ***Consistent:*** Blockchain data is complete, consistent, timely, accurate, and widely available.
- ***Persistence:*** No invalid transactions are admitted. It is nearly impossible to delete or roll back transactions once they are included in the blockchain. Cryptographically, the blocks created are sealed in the chain. It is impossible to delete, edit or copy already created blocks and put them on the network. This ensures a high level of robustness and trust.
- ***Anonymity:*** Each user can interact with the blockchain with a generated address, which does not reveal the real identity of the user.

Principles of blockchain

- ***Integrity:*** Blockchain technology uses a distributed verification system for transactions that does not centralise power. Integrity is required during the decision rights of blockchain users and a network's incentive structures. The use of complex algorithms and consensus among users ensures that once agreed upon, there is no tampering with transaction data. Data stored on blockchain acts as a single version of truth for all parties involved, reducing the risk of fraud.
- ***Decentralisation:*** Blockchain distributes power among different members. This means no single person or system can ever tamper with it. It is robust and resistant to attacks and fraud due to a distributed power system.
- ***Peer-to-peer:*** All blockchain transactions are peer-to-peer. This means that at least two parties are involved. The best way to understand this is to compare a standard credit card transaction with the first practical use of blockchain. The benefits of the peer-to-peer aspects of the blockchain are that the two people involved in a transaction can both verify if the transaction is correct, without the need for any other party to be involved.
- ***Cryptographic system:*** Blockchain lends its cryptography a level of authenticity and integrity.
- ***Equal rights:*** Rights are reserved in a transparent manner. Blockchain users know exactly what rights they are entitled to and that they are equal and fair in the best way possible.

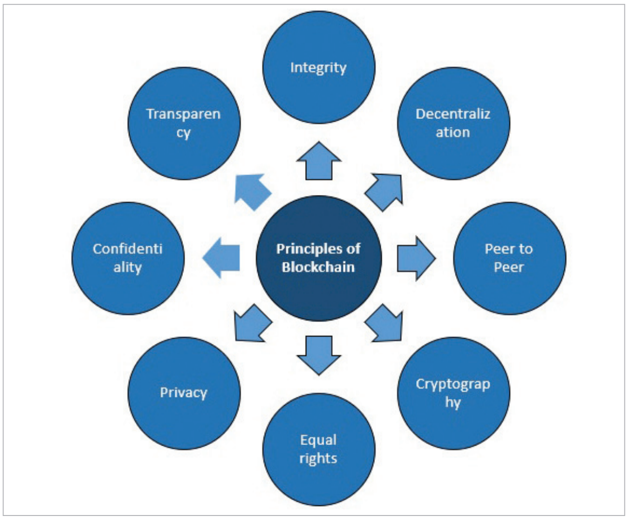

Figure 1: Principles of blockchain

- ***Privacy:*** Blockchain establishes a separate virtual identity for its users to protect their privacy, both online and offline. The identity uses a series of numbers, much like code, that are difficult to profile. Hence, blockchain fosters trust by eliminating the need for any real (or seemingly real) identity. It protects the identity of its users so that they can participate in the digital economy freely and fairly with integrity.
- ***Confidentiality:*** Blockchain provides confidentiality by enabling users of a ledger to see authorised transactions only.
- ***Transparency:*** Blockchain shares the ledger to all nodes and uses consensus algorithms to reach consensus among them. Consensus algorithms also ensure the ordering and execution of the transactions.

Figure 2 shows the logical application architecture of the blockchain system with key components and layers.

Blockchain services layer acts as a gateway for reaching out to the blockchain infrastructure. It consists of various services related to identity management, key management, cryptography, machine learning (ML) and business intelligence (BI). Identity, key, and cryptographic services are critical for data integrity. BI and ML are used for analysis and reporting purposes.

Distributed ledger stack consists of the core distributed ledger technology stack. Blockchain building, transaction execution and consensus happen in this layer. The components in this stack may vary depending on the distributed ledger product chosen. However, the common sub-components are consensus algorithms, data storage, and transactions management and other relevant services.